Hadoop HDFS Practice Commands

Part 1: Start Hadoop Services in Ubuntu

1. Open Terminal in Ubuntu

Press:

Ctrl + Alt + T

2. Go to Hadoop sbin Folder

cd /home/hadoop/hadoop/sbin

3. Start HDFS

./start-dfs.sh

4. Start YARN

./start-yarn.sh

5. Check Running Hadoop Daemons

jps

Part 2: Practice HDFS Commands

1. Create Directory (mkdir)

Create directory in HDFS:

hdfs dfs -mkdir /abc

Check directory:

hadoop fs -ls /

2. Upload File Using put

First create a text file.

Create file:

gedit sample.txt

Paste the following text into the file:

hi how are you
how is your job
how is your family
how is your brother
how is your sister

Save the file.

Upload file to HDFS:

hdfs dfs -put sample.txt /abc/WC.txt

Check:

hdfs dfs -ls /abc

3. Display File (cat)

hdfs dfs -cat /abc/WC.txt

4. copyFromLocal

hdfs dfs -copyFromLocal sample.txt /abc/WC2.txt

Check file content:

hdfs dfs -cat /abc/WC2.txt

5. get (Download from HDFS)

hdfs dfs -get /abc/WC.txt ~/Downloads/WWC.txt

6. getmerge

Merge two HDFS files into one local file:

hdfs dfs -getmerge /abc/WC.txt /abc/WC2.txt ~/Desktop/Merge.txt

7. copyToLocal

hdfs dfs -copyToLocal /abc/WC.txt ~/Desktop

8. Move Directory

hadoop fs -mv /abc /FFF

Check:

hadoop fs -ls /FFF

9. Copy Directory

hadoop fs -cp /FFF /LLL

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hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC: $ cd /home/hadoop/hadoop/sbin
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ ./start-dfs.sh
Starting namenodes on [localhost]
localhost: namenode is running as process 13038.  Stop it first and ensure /tmp/hadoop-hadoop-namenode.pid file is empty before retry.
Starting datanodes
localhost: datanode is running as process 13228.  Stop it first and ensure /tmp/hadoop-hadoop-datanode.pid file is empty before retry.
Starting secondary namenodes [bmsce-HP-Elite-Tower-600-G9-Desktop-PC]
bmsce-HP-Elite-Tower-600-G9-Desktop-PC: secondarynamenode is running as process 13538.  Stop it first and ensure /tmp/hadoop-hadoop-secondarynamenode.pid file is empty before retry.
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ ./start-yarn.sh
Starting resourcemanager
resourcemanager is running as process 13877.  Stop it first and ensure /tmp/hadoop-hadoop-resourcemanager.pid file is empty before retry.
Starting nodemanagers
localhost: nodemanager is running as process 14043.  Stop it first and ensure /tmp/hadoop-hadoop-nodemanager.pid file is empty before retry.
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ jps
13538 SecondaryNameNode
15538 Jps
13877 ResourceManager
14043 NodeManager
13228 DataNode
13038 NameNode
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -mkdir /abc
mkdir: `/abc': File exists
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hadoop fs -ls
ls: `.': No such file or directory
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hadoop fs -ls /
Found 11 items
drwxr-xr-x   - hadoop supergroup          0 2025-04-16 11:43 /FFF
drwxr-xr-x   - hadoop supergroup          0 2025-04-16 11:56 /LLL
drwxr-xr-x   - hadoop supergroup          0 2026-02-25 15:54 /abc
drwxr-xr-x   - hadoop supergroup          0 2026-02-25 15:55 /hadoop_lab
drwxr-xr-x   - hadoop supergroup          0 2026-02-25 16:05 /inputbda
drwxr-xr-x   - hadoop supergroup          0 2026-02-25 16:06 /outputbda
drwxr-xr-x   - hadoop supergroup          0 2024-09-10 12:56 /pqr
drwxr-xr-x   - hadoop supergroup          0 2025-05-07 12:26 /rgs
drwxr-xr-x   - hadoop supergroup          0 2026-02-25 15:50 /test
drwxr-xr-x   - hadoop supergroup          0 2025-05-27 15:31 /user
drwxr-xr-x   - hadoop supergroup          0 2024-09-10 13:19 /xyz
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ gedit sample.txt

hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -put sample.txt /abc/WC.txt
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -ls /abc
Found 3 items
-rw-r--r--   1 hadoop supergroup         89 2026-03-06 15:41 /abc/WC.txt
-rw-r--r--   1 hadoop supergroup         17 2026-02-25 15:53 /abc/file.txt
-rw-r--r--   1 hadoop supergroup         17 2026-02-25 15:54 /abc/file_cp_local.txt
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -cat /abc/WC.txt
hi how are you
how is your job
how is your family
how is your brother
how is your sister
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -copyFromLocal sample.txt /abc/WC2.txt
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -cat /abc/WC2.txt
hi how are you
how is your job
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14043 NodeManager
13228 DataNode
13038 NameNode
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -mkdir /abc
mkdir: `/abc': File exists
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hadoop fs -ls
ls: `.': No such file or directory
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hadoop fs -ls /
Found 11 items
drwxr-xr-x   - hadoop supergroup          0 2025-04-16 11:43 /FFF
drwxr-xr-x   - hadoop supergroup          0 2025-04-16 11:56 /LLL
drwxr-xr-x   - hadoop supergroup          0 2026-02-25 15:54 /abc
drwxr-xr-x   - hadoop supergroup          0 2026-02-25 15:55 /hadoop_lab
drwxr-xr-x   - hadoop supergroup          0 2026-02-25 16:05 /inputbda
drwxr-xr-x   - hadoop supergroup          0 2026-02-25 16:06 /outputbda
drwxr-xr-x   - hadoop supergroup          0 2024-09-10 12:56 /pqr
drwxr-xr-x   - hadoop supergroup          0 2025-05-07 12:26 /rgs
drwxr-xr-x   - hadoop supergroup          0 2026-02-25 15:50 /test
drwxr-xr-x   - hadoop supergroup          0 2025-05-27 15:31 /user
drwxr-xr-x   - hadoop supergroup          0 2024-09-10 13:19 /xyz
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ gedit sample.txt

hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -put sample.txt /abc/WC.txt
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -ls /abc
Found 3 items
-rw-r--r--   1 hadoop supergroup         89 2026-03-06 15:41 /abc/WC.txt
-rw-r--r--   1 hadoop supergroup         17 2026-02-25 15:53 /abc/file.txt
-rw-r--r--   1 hadoop supergroup         17 2026-02-25 15:54 /abc/file_cp_local.txt
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -cat /abc/WC.txt
hi how are you
how is your job
how is your family
how is your brother
how is your sister
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -copyFromLocal sample.txt /abc/WC2.txt
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -cat /abc/WC2.txt
hi how are you
how is your job
how is your family
how is your brother
how is your sister
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -get /abc/WC.txt ~/Downloads/WWC.txt
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -getmerge /abc/WC.txt /abc/WC.txt /abc/WC2.txt ~Desktop/Merge.txt
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hdfs dfs -copyToLocal /abc/WC.txt ~/Desktop
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hadoop fs -mv /abc /FFF
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hadoop fs -ls /FFF
Found 5 items
-rw-r--r--   1 hadoop supergroup         48 2025-04-16 11:43 /FFF/WC.txt
drwxr-xr-x   - hadoop supergroup          0 2026-03-06 15:42 /FFF/abc
-rw-r--r--   1 hadoop supergroup         48 2024-09-10 13:02 /FFF/wc.txt
-rw-r--r--   1 hadoop supergroup         48 2024-09-10 13:02 /FFF/wc1.txt
-rw-r--r--   1 hadoop supergroup         96 2024-09-10 13:22 /FFF/wc3.txt
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ hadoop fs -cp /FFF /LLL
hadoop@bmsce-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$
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